

Correct Marks : 3

Question Label : Short Answer Question

If we want to implement the minimized expression using only 2-input NOR gates, what is the minimum number of NOR gates required, given inputs A, B, C, and D?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

1

Question Number : 91 Question Id : 6406531337372 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

If we want to design the given minterm using 2:1 multiplexers, what is the minimum number of multiplexers required?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

2

Sw Engg

Section Id :	64065394926
Section Number :	6
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	23
Number of Questions to be attempted :	23
Section Marks :	100
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	No
Section Maximum Duration :	0
Section Minimum Duration :	0

Section Time In :Minutes

Maximum Instruction Time :0

Sub-Section Number :1

Sub-Section Id :640653217618

Question Shuffling Allowed :No

Question Number : 92 Question Id : 6406531337373 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : SOFTWARE ENGINEERING (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406534468628. ✓ YES
6406534468629. ✗ NO

Sub-Section Number :2

Sub-Section Id :640653217619

Question Shuffling Allowed :Yes

Question Number : 93 Question Id : 6406531337374 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Match the type of client with the corresponding example in a university software ecosystem.

Type of Client	Example
A. External User	1. Faculty dashboard to manage course materials
B. Internal to your Company	2. Library system syncing with a third-party digital archive
C. Another Software	3. Online application portal used by prospective students

Options :

6406534468630. ✗ A-1, B-2, C-3
6406534468631. ✓ A-3, B-1, C-2
6406534468632. ✗ A-2, B-3, C-1
6406534468633. ✗ A-1, B-3, C-2

Question Number : 94 Question Id : 6406531337376 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

You're tasked with building a digital wallet feature for an e-commerce platform. Your team suggests directly looking into how similar apps like Paytm and PhonePe work.

Which of the following best describes the activity being performed?

Options :

6406534468638. ✖ Software Testing

6406534468639. ✔ Requirements Analysis

6406534468640. ✖ Code Optimization

6406534468641. ✖ Alpha Testing

Question Number : 95 Question Id : 6406531337377 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

You are developing an online education platform. Professors use the system to upload materials, students access the content, and the university's accreditation body occasionally reviews course reports generated from the system.

Which of the following correctly categorizes the stakeholders?

Options :

6406534468642. ✖ Professors – Primary users; Students – Secondary users; Accreditation body – Tertiary users

6406534468643. ✖ Professors – Secondary users; Students – Primary users; Accreditation body – Tertiary users

6406534468644. ✔ Professors – Primary users; Students – Primary users; Accreditation body – Tertiary users

6406534468645. ✖ Professors – Tertiary users; Students – Secondary users; Accreditation body – Primary users

Question Number : 96 Question Id : 6406531337378 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

One stakeholder wants user passwords to expire every 30 days. Another insists users must not be forced to change passwords more than once a year.

What kind of requirement issue is this?

Options :

6406534468646. ✖ Incomplete

6406534468647. ✖ Ambiguous

6406534468648. ✔ Inconsistent

6406534468649. ✖ Implicit

Question Number : 97 Question Id : 6406531337379 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

You are part of a software team following Agile methodology. A client provides a user story:

As a seller, I want to view product reviews so that I can improve my inventory.

However, during implementation, the team realizes the user also wants email alerts for 1-star reviews.

Which of the following Agile principles best explains the incorporation of this new requirement?

Options :

6406534468650. ✖ Working software is the primary measure of progress

6406534468651. ✔ Customer collaboration over contract negotiation

6406534468652. ✖ Processes and tools over individuals and interactions

6406534468653. ✖ Following a plan over responding to change

Question Number : 98 Question Id : 6406531337380 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

While designing the seller portal's financial features, your team needs to comply with local banking regulations on fund transfers. You refer to official banking policies and competitor portal FAQs.

Which requirement gathering technique does this illustrate?

Options :

6406534468654. ✖ Focus group

6406534468655. ✖ Observation

6406534468656. ✔ Documentation

6406534468657. ✖ Interview

Question Number : 99 Question Id : 6406531337381 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

The following user story is proposed:

As a seller, I want to get suggestions on bundling products based on previous buyer behavior, so that I can increase cross-selling opportunities.

The team agrees it's a valuable feature, but the Scrum Master notes that the story lacks a deadline or timeframe.

Which SMART criterion is lacking?

Options :

6406534468658. ✖ Measurable

6406534468659. ✖ Specific

6406534468660. ✔ Time-boxed

6406534468661. ✖ Relevant

Question Number : 100 Question Id : 6406531337382 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A startup is developing an app for food delivery. They've created clickable images of app screens using PowerPoint and ask users to try placing a test order.

What kind of prototype is being used?

Options :

6406534468662. ✖ Storyboard

6406534468663. ✖ Paper prototype

6406534468664. ✔ Digital mock-up

6406534468665. ✖ Interactive prototype

Question Number : 101 Question Id : 6406531337383 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A Scrum team has assigned point values to each user story. In their first sprint, they completed 2 stories totaling 5 points.

What is the estimated velocity of the team?

Options :

6406534468666. ✖ 2

6406534468667. ✔ 5

6406534468668. ✖ 2.5

6406534468669. ✖ 10

Question Number : 102 Question Id : 6406531337384 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A company develops a mobile game for low-end devices. During design, the team restricts image resolution and optimizes memory usage by reusing textures.

Which quality of good design is being prioritized here?

Options :

6406534468670. ✖ Understandability

6406534468671. ✖ Maintainability

6406534468672. ✔ Efficiency

6406534468673. ✖ Scalability

Question Number : 103 Question Id : 6406531337386 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A programmer is working with a third-party Python API for image processing. Their code compiles, but the output image is not what they expected. They cannot see or change the library's internal code.

Which debugging technique should they use?

Options :

6406534468678. ✖ Step into each function using PDB

6406534468679. ✖ Forward debugging to trace internal logic

6406534468680. ✔ Black box debugging using documentation and examples

6406534468681. ✖ Dump and diff comparison between images

Question Number : 104 Question Id : 6406531337387 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Consider the following Python function:

```
def process_list(nums):
    total = 0
    for num in nums:

        if num % 2 == 0:
            total += num
        else:
            total -= num
    if total > 0:
        return "Positive sum"
    return "Non-positive sum"
```

What is the cyclomatic complexity of the above function?

Options :

6406534468682. ✖ 2

6406534468683. ✖ 3

6406534468684. ✔ 4

6406534468685. ✖ 5

Question Number : 105 Question Id : 6406531337388 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A file-sharing system allows users to both download and upload files from / to other users without relying on any central server. Each user system shares a part of their storage. What architecture is being described?

Options :

6406534468686. ✖ Client-Server

6406534468687. ✖ Model-View-Controller

6406534468688. ✖ Pipe and Filter

6406534468689. ✔ Peer-to-Peer (P2P)

Question Number : 106 Question Id : 6406531337389 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

In an e-commerce seller portal, a *DeliveryProcessing* interface includes the following methods:

- *place_order()*
- *notify_seller()*
- *package_order()*
- *deliver_to_address()*

Two types of products are supported:

- Physical products (e.g., clothes, books)
- Digital products (e.g., eBooks, software licenses)

The *DigitalOrderProcessor* class is supposed to process only digital orders. However, due to the design of *DeliveryProcessing*, it is forced to implement all methods, including *package_order()* and *deliver_to_address()*, which are not required for digital products.

Which SOLID principle is being violated?

Options :

6406534468690. ✖ Single responsibility principle

6406534468691. ✔ Interface segregation principle

6406534468692. ✖ Open-closed principle

6406534468693. ✖ Liskov substitution principle

Question Number : 107 Question Id : 6406531337390 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A startup is building an e-learning platform that allows instructors to create and publish new courses. To publish a course, the system performs the following operations:

1. Verifies the instructor's identity and permissions
2. Checks the course content for plagiarism
3. Converts uploaded videos into streaming formats
4. Updates the course catalog and sends email notifications to subscribers

Initially, the course publishing feature was implemented by calling individual services directly from the UI layer. However, developers found this process tedious and error-prone, as it required manually orchestrating multiple steps and handling different exceptions.

To streamline this, the development team created a *CoursePublisher* class that provides a simple method *publishCourse(course)*, which internally handles all the above tasks in the correct order.

Which design pattern does this solution illustrate?

Options :

- 6406534468694. ✖ Builder design pattern
- 6406534468695. ✖ Adapter Pattern
- 6406534468696. ✔ Facade Pattern
- 6406534468697. ✖ Strategy design pattern

Sub-Section Number :	3
Sub-Section Id :	640653217620
Question Shuffling Allowed :	Yes

Question Number : 108 Question Id : 6406531337375 Question Type : MCQ

Correct Marks : 5

Question Label : Multiple Choice Question

Your team developed a system over 12 months using a strict sequential process. After the final release, the client expresses dissatisfaction due to misunderstood requirements. What was the likely development model?

Options :

- 6406534468634. ✖ Agile
- 6406534468635. ✖ Spiral
- 6406534468636. ✔ Waterfall
- 6406534468637. ✖ Prototype

Question Number : 109 Question Id : 6406531337385 Question Type : MCQ

Correct Marks : 5

Question Label : Multiple Choice Question

In an e-commerce system, the *SellerModule* calls a function in the *InventoryModule* using a variable *operationFlag* to decide whether to view, add, or delete a product. Which type of coupling is present?

Options :

- 6406534468674. ✓ Control Coupling
- 6406534468675. ✗ Content Coupling
- 6406534468676. ✗ Data Coupling
- 6406534468677. ✗ Common Coupling

Sub-Section Number : 4
Sub-Section Id : 640653217621
Question Shuffling Allowed : Yes

Question Number : 110 Question Id : 6406531337391 Question Type : MSQ
Correct Marks : 5 Max. Selectable Options : 0

Question Label : Multiple Select Question

In an online seller portal, the *Seller* class uses the *Inventory* class's *viewInventory(sellerID)* method to fetch the list of products. Additionally, the *Inventory* class implements an interface *InventoryInterface* that defines standard inventory operations.

Which of the following relationships can be identified from this scenario?

Options :

- 6406534468698. ✗ *Seller* and *Inventory* have an inheritance relationship
- 6406534468699. ✓ *Seller* and *Inventory* have an association relationship
- 6406534468700. ✓ *Inventory* and *InventoryInterface* have a dependency relationship
- 6406534468701. ✗ *Seller* and *InventoryInterface* have a composition relationship

Question Number : 111 Question Id : 6406531337393 Question Type : MSQ
Correct Marks : 5 Max. Selectable Options : 0

Question Label : Multiple Select Question

A team is designing a new tax filing software for first-time users. Which of the following features would most likely improve the learnability of their system?

Options :

- 6406534468706. ✓ Step-by-step wizard that guides users through each section
- 6406534468707. ✗ Ability to undo recent entries using a shortcut
- 6406534468708. ✓ Templates for common user profiles (e.g., salaried, freelancer)
- 6406534468709. ✗ Strong password requirement with special characters and numbers

Sub-Section Number : 5
Sub-Section Id : 640653217622
Question Shuffling Allowed : Yes

Question Number : 112 Question Id : 6406531337392 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

In a distributed software development environment, why is it important to define **interfaces** clearly before implementing components?

Options :

6406534468702. ✖ It ensures that all teams use the same programming language for implementation.

6406534468703. ✔ It helps maintain consistent communication between components developed by different team members.

6406534468704. ✔ It allows the internal implementation of a component to change without affecting others.

6406534468705. ✖ It eliminates the need for documentation, since the code is self-explanatory.

Sub-Section Number : 6

Sub-Section Id : 640653217623

Question Shuffling Allowed : No

Question Id : 6406531337394 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (113 to 114)

Question Label : Comprehension

A module in your project depends on a third-party payment gateway. During testing, you discover it is not functioning properly due to a known bug in the vendor's code.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 113 Question Id : 6406531337395 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Based on the scenario, which category does this risk most appropriately fall under?

Options :

6406534468710. ✖ Business risk

6406534468711. ✔ Technical risk

6406534468712. ✖ Project risk

6406534468713. ✖ Legal risk

Question Number : 114 Question Id : 6406531337396 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

What is the most suitable way to mitigate the risk described in the scenario?

Options :

6406534468714. ✖ Conduct user feedback surveys

6406534468715. ✖ Increase the project budget to cover delays

6406534468716. ✔ Benchmark and conduct regular inspections of third-party components

6406534468717. ✖ Ask developers to avoid using third-party modules altogether

Question Id : 6406531337397 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (115 to 116)

Question Label : Comprehension

You are estimating the effort required to develop an internal business dashboard for a mid-sized company. The system includes analytics, reporting, and user access modules.

The team has a mix of experienced and relatively new developers, and the organization has built a few similar systems before.

The estimated size of the project is 6 KLOC.

The values for the constants in the COCOMO model for the different types of projects are as follows:

- Organic project: $a = 2.4$, $b = 1.05$
- Semi-detached project: $a = 3.0$, $b = 1.12$
- Embedded project: $a = 3.6$, $b = 1.20$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 115 Question Id : 6406531337398 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

What is the estimated initial effort for the dashboard system?

Options :

6406534468718. ✖ 18.6 person-months

6406534468719. ✔ 22.3 person-months

6406534468720. ✖ 23.5 person-months

6406534468721. ✖ 25.0 person-months

Question Number : 116 Question Id : 6406531337399 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question